

HDOC Day-42

Menu-Driven String Operations in C

Problem Statement

Write a menu-driven C program with separate user-defined functions to remove vowels, find the first repeating lowercase alphabet, and test whether two strings are anagrams.

Algorithm

1. Display the menu repeatedly until Exit is chosen.
2. Choice 1: read a string; scan every character; copy only non-vowels into a second string, then display it.
3. Choice 2: read a string; keep a frequency/seen array for lowercase a-z; report the first lowercase character encountered for the second time, or report none.
4. Choice 3: read two strings; count character occurrences after converting letters to lowercase and ignoring whitespace; compare frequencies and report whether they are anagrams.
5. Choice 4: exit; otherwise display an invalid-choice message.

Test Case Table

TC	Choice	Input	Expected Result
1	1	Education	After removing vowels: dctn
2	1	AEIOU	After removing vowels: (empty)
3	2	banana	First repeating lowercase alphabet: a
4	2	abcABC	No repeating lowercase alphabet found.
5	3	Listen / Silent	Strings are anagrams.
6	3	hello / world	Strings are not anagrams.
7	9	Invalid menu choice	Invalid choice. Enter 1 to 4.
8	4	Exit	Exiting program.

Anagram comparison ignores whitespace and letter case; other characters are counted. Only lowercase a-z characters are considered for the repeating-letter test.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 500

void readLine(char s[], int capacity) {
    if (fgets(s, capacity, stdin) == NULL) { s[0] = '\0'; return; }
    s[strcspn(s, "\n")] = '\0';
}

void removeVowels(void) {
    char s[MAX], result[MAX]; int i, j = 0;
```

```

printf("Enter a string: "); readLine(s, MAX);
for (i = 0; s[i] != '\0'; i++) {
    char c = (char)tolower((unsigned char)s[i]);
    if (c != 'a' && c != 'e' && c != 'i' && c != 'o' && c != 'u') result[j++] = s[i];
}
result[j] = '\0'; printf("After removing vowels: %s\n", result);
}

void firstRepeatingLowercase(void) {
    char s[MAX]; int seen[26] = {0}, i;
    printf("Enter a string: "); readLine(s, MAX);
    for (i = 0; s[i] != '\0'; i++) {
        if (s[i] >= 'a' && s[i] <= 'z') {
            int index = s[i] - 'a';
            if (seen[index]) { printf("First repeating lowercase alphabet: %c\n", s[i]); return; }
            seen[index] = 1;
        }
    }
    printf("No repeating lowercase alphabet found.\n");
}

void checkAnagrams(void) {
    char a[MAX], b[MAX]; int freq[256] = {0}, i;
    printf("Enter first string: "); readLine(a, MAX);
    printf("Enter second string: "); readLine(b, MAX);
    /* Ignore spaces and letter case; other characters are compared. */
    for (i = 0; a[i] != '\0'; i++)
        if (!isspace((unsigned char)a[i])) freq[(unsigned char)tolower((unsigned char)a[i])]++;
    for (i = 0; b[i] != '\0'; i++)
        if (!isspace((unsigned char)b[i])) freq[(unsigned char)tolower((unsigned char)b[i])]--;
    for (i = 0; i < 256; i++)
        if (freq[i] != 0) { printf("Strings are not anagrams.\n"); return; }
    printf("Strings are anagrams.\n");
}

int main(void) {
    char line[64]; int choice;
    while (1) {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Remove vowels\n2. First repeating lowercase alphabet\n3. Check anagrams\n4. Exit\nEnter choice: ");
        readLine(line, sizeof line);
        if (sscanf(line, "%d", &choice) != 1) { puts("Invalid choice. Enter 1 to 4."); continue; }
        switch (choice) {
            case 1: removeVowels(); break;
            case 2: firstRepeatingLowercase(); break;
            case 3: checkAnagrams(); break;
            case 4: puts("Exiting program."); return 0;
            default: puts("Invalid choice. Enter 1 to 4.");
        }
    }
}

```

Compilation, Execution and Test Case Screenshots

Compilation and Execution

```
(kali㉿kali)-[~/Desktop]
$ vi hdoc42.c

(kali㉿kali)-[~/Desktop]
$ gcc hdoc42.c -o hdoc42

(kali㉿kali)-[~/Desktop]
$ ./hdoc42
```

Test Case 1

Input: Education | Expected: After removing vowels: dctn

```
4. EXIT
Enter your choice: 1
Enter a string: Education
String after removing vowels: dctn
```

Test Case 2

Input: AEIOU | Expected: After removing vowels: (empty)

```
4. EXIT
Enter your choice: 1
Enter a string: AEIOU
String after removing vowels:
```

Test Case 3

Input: banana | Expected: First repeating lowercase alphabet: a

```
4. EXIT
Enter your choice: 2
Enter a string: banana
First repeating lowercase alphabet: a
```

Test Case 4

Input: abcABC | Expected: No repeating lowercase alphabet found.

```
4. EXIT
Enter your choice: 2
Enter a string: abcABC
No repeating lowercase alphabet found.
```

Test Case 5

Input: Listen / Silent | Expected: Strings are anagrams.

```
4. EXIT
Enter your choice: 3
Enter first string: Listen
Enter second string: Silent
The two strings are anagrams.
```

Test Case 6

Input: hello / world | Expected: Strings are not anagrams.

```
Enter your choice: 3
Enter first string: hello
Enter second string: world
The two strings are not anagrams.
```

Test Case 7

Input: Invalid menu choice | Expected: Invalid choice. Enter 1 to 4.

```
4. EXIT
Enter your choice: 9
Invalid choice. Enter 1 to 4.
```

Test Case 8

Input: Exit | Expected: Exiting program.

```
4. EXIT
Enter your choice: 4
Exiting program.
```